

HR WORKFORCE ATTRITION ANALYSIS

ABSTRACT:

Employee attrition is a major concern for organizations as it affects productivity, operational efficiency, and recruitment costs. This project analyzes HR workforce data to identify the factors influencing employee turnover. Data cleaning, preprocessing, and exploratory data analysis were performed using Python, while Power BI was used to develop interactive dashboards. The analysis focuses on employee satisfaction, salary levels, promotions, overtime, and departmental performance. The insights generated help organizations understand attrition patterns and support data-driven HR decision-making to improve employee retention.

INTRODUCTION:

Human Resource Analytics plays a vital role in understanding employee behavior and improving workforce management. Employee attrition can negatively impact organizational performance by increasing hiring costs and reducing productivity. The objective of this project is to analyze employee data and identify the key factors contributing to attrition. Using Python for data analysis and Power BI for visualization, the project examines employee satisfaction, work conditions, salary structure, promotions, and risk categories. The findings provide valuable insights that can help HR teams develop effective strategies to enhance employee engagement and retention.

DATASET DESCRIPTION:

The dataset contains employee information related to:

- Satisfaction Level

- Last Evaluation
- Number of Projects
- Average Monthly Hours
- Time Spent in Company
- Work Accident History
- Promotion Status
- Department
- Salary Level
- Attrition Status

DATA PREPARATION:

Handling Missing Values:

The dataset was checked for missing values to ensure data quality and reliability. During the analysis, no missing values were found in any of the columns. Therefore, no imputation or replacement techniques were required. All records were retained and used for further analysis without any modification.

Handling Duplicate Records:

The dataset was examined for duplicate records to prevent inaccurate employee counts and misleading attrition insights. During the analysis, 3,008 duplicate records were identified.

These duplicate entries were removed using appropriate data cleaning techniques to ensure that each employee was represented only once in the dataset. As a result, the dataset size was reduced from 14,999

records to 11,991 unique employee records, which were used for subsequent analysis and dashboard development.

KPI SUMMARY:

KPI	Value
Total Employees	11,991
Attrition Rate	16.60%
Retention Rate	83.40%
Average Tenure	Calculated from dashboard
Satisfaction Score	Calculated from dashboard
High Risk Employees	Calculated from dashboard

ANALYTICS INVESTIGATION:

1. Department Analysis:

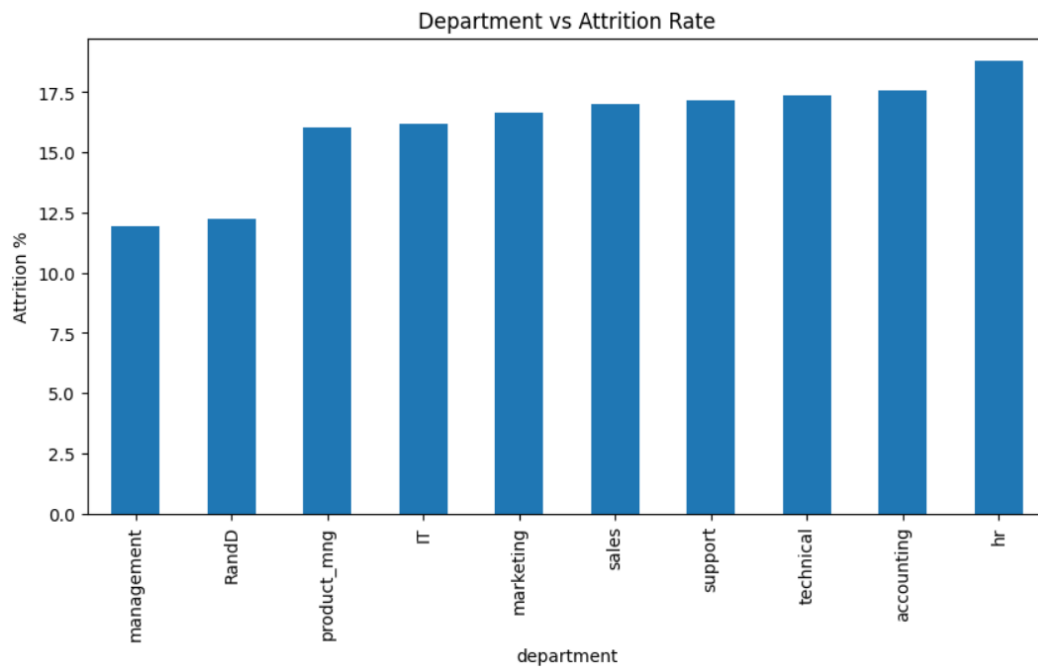
Highest attrition departments:

1. HR – 18.80%
2. Accounting – 17.55%
3. Technical – 17.38%
4. Support – 17.13%

Lowest attrition departments:

1. Management – 11.93%
2. R&D – 12.25%

This indicates that HR, Accounting, Technical, and Support departments experience greater employee turnover and may require targeted retention strategies.

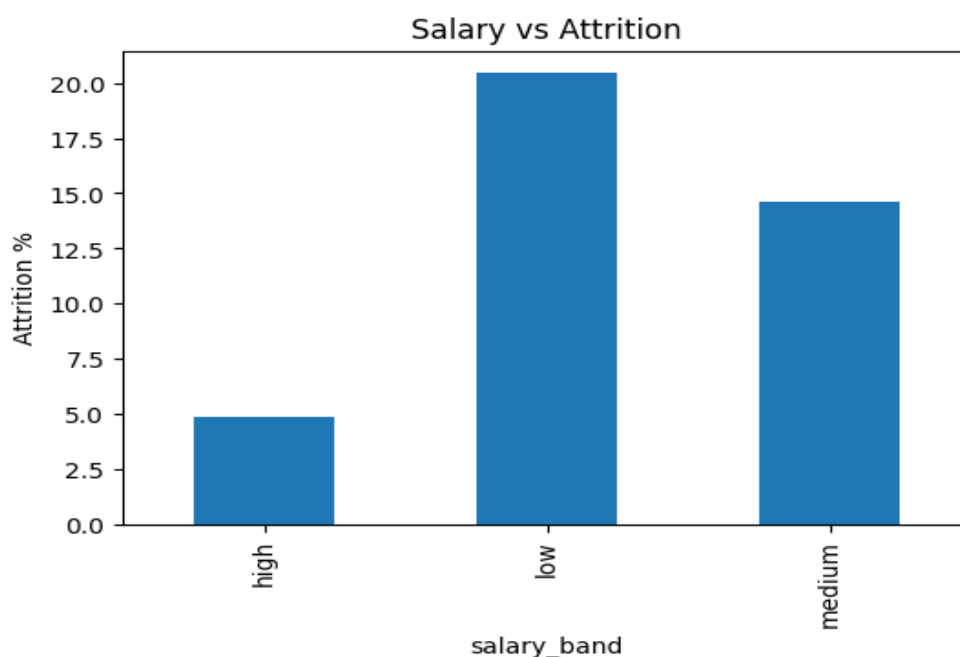


Salary Analysis:

Attrition Rate by Salary:

- Low Salary: 20.45%
- Medium Salary: 14.62%
- High Salary: 4.85%

Employees receiving low salaries leave the organization significantly more often than employees receiving higher salaries.

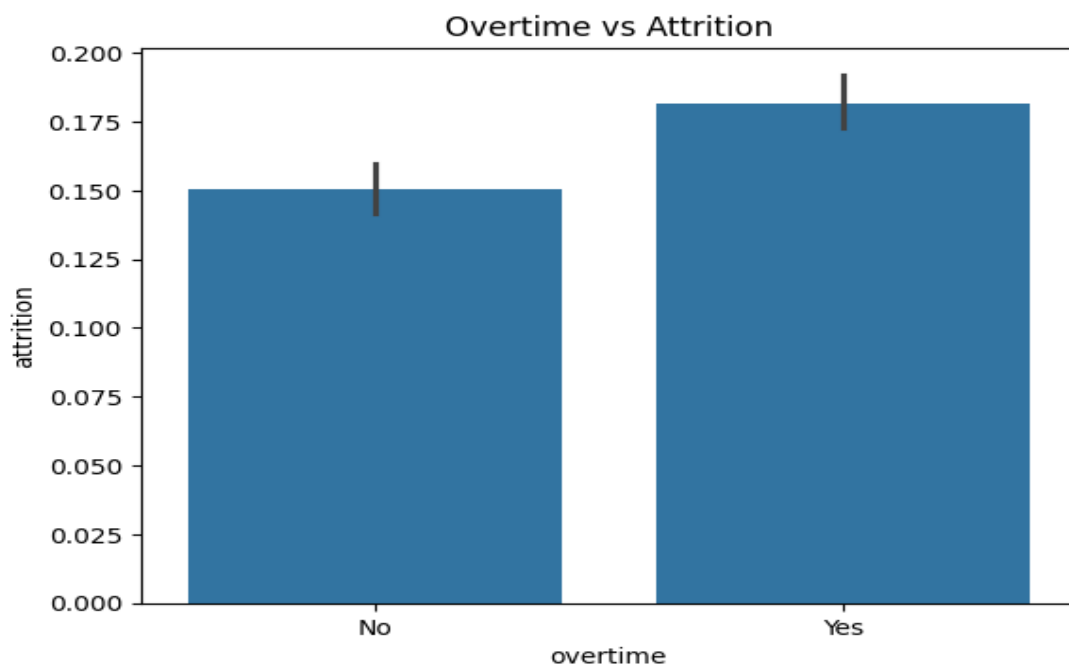


Overtime Analysis:

Attrition Rate:

- Overtime Employees: 18.18%
- Non-Overtime Employees: 15.05%

Employees working overtime are more likely to leave the organization.

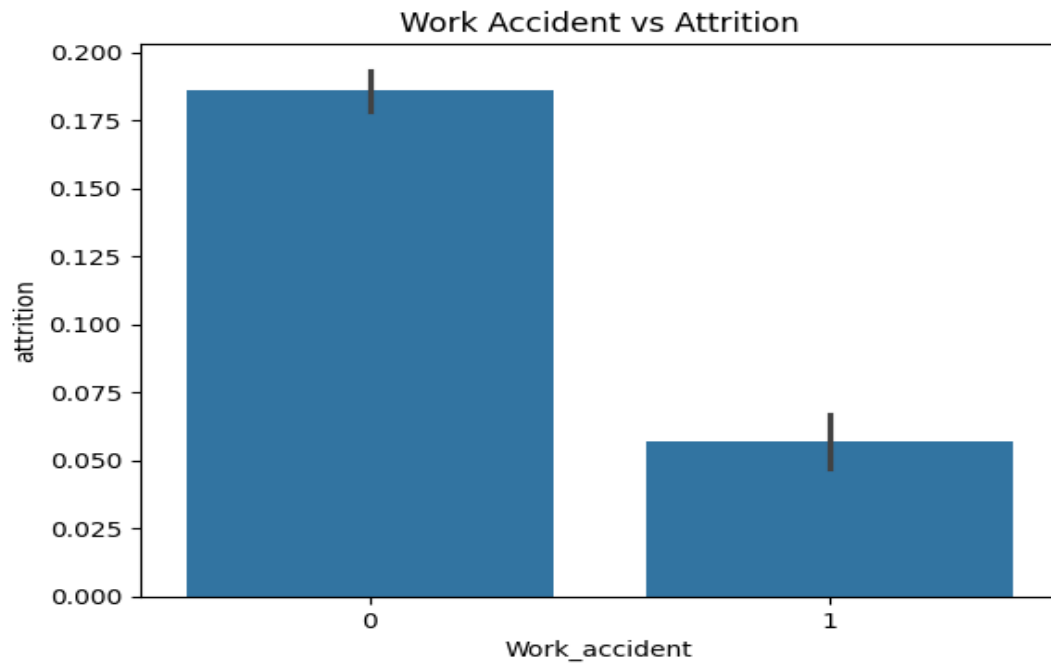


Work Accident Analysis:

Attrition Rate:

- No Work Accident: 18.60%
- Work Accident: 5.68%

Employees who experienced work accidents show lower attrition compared to others, possibly because they receive additional organizational support and benefits.

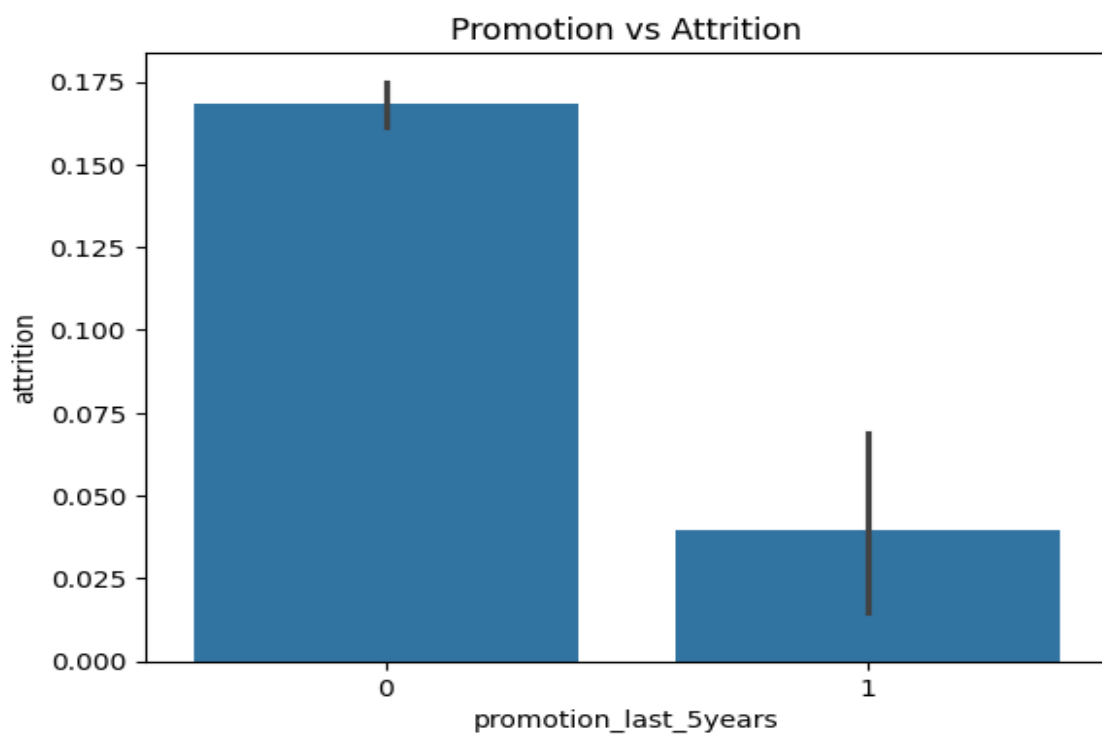


Promotion Analysis:

Attrition Rate:

- Not Promoted: 16.82%
- Promoted: 3.94%

Employees who receive promotions are substantially more likely to stay with the organization.

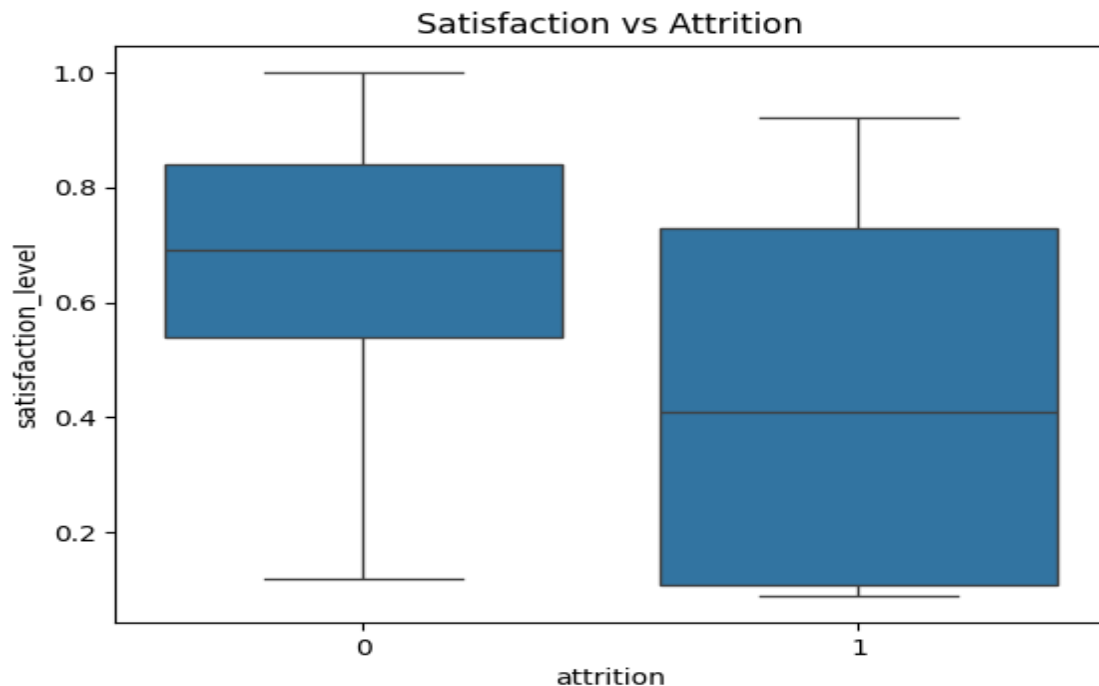


Satisfaction Analysis:

Correlation between Satisfaction and Attrition:

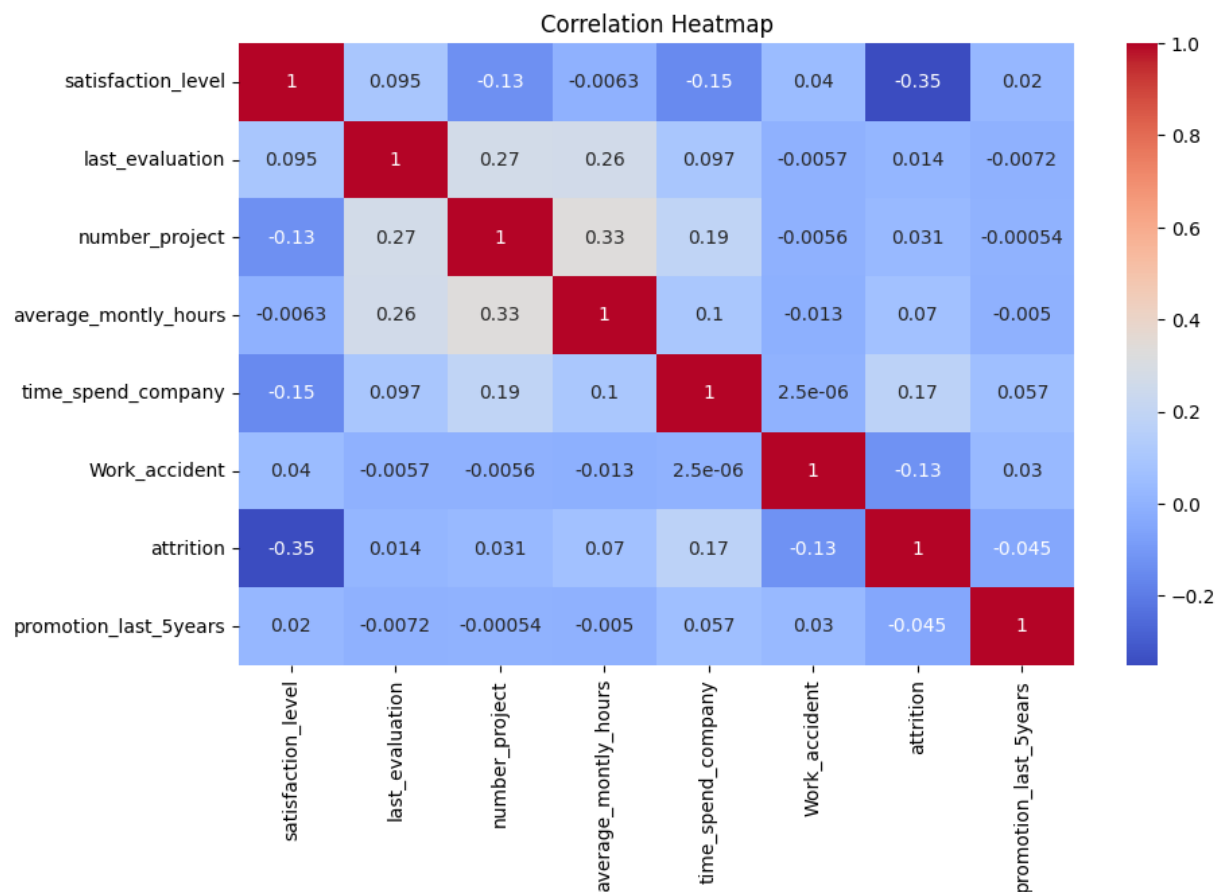
- Correlation = -0.35

This negative correlation indicates that as employee satisfaction decreases, attrition increases.



Correlation Heatmap:

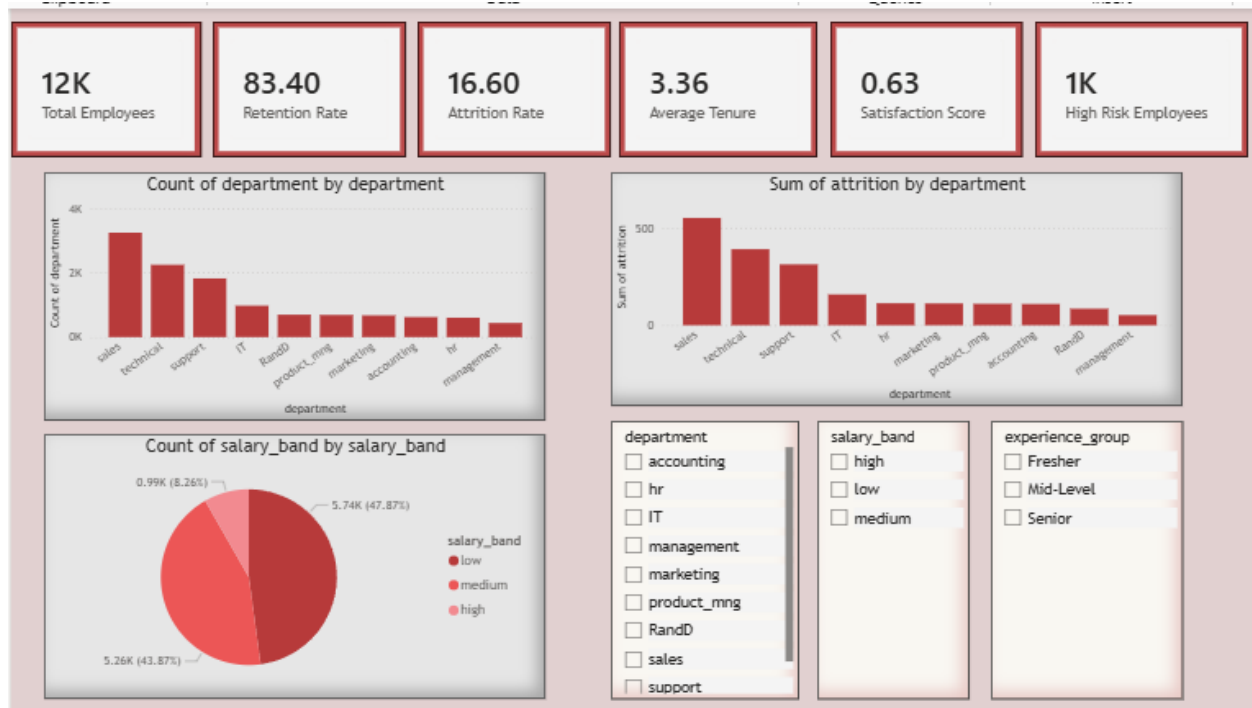
The correlation heatmap helped determine which factors are associated with employee attrition, such as satisfaction level, monthly working hours, projects, and promotions. This made it easier to identify important variables affecting employee retention.



DASHBOARD DEVELOPMENT:

Executive Overview Dashboard:

The Executive Overview Dashboard presents a summary of the organization's workforce through key performance indicators such as Total Employees, Attrition Rate, Retention Rate, Average Tenure, Employee Satisfaction Score, and High-Risk Employees. This page provides a quick assessment of the overall workforce condition and highlights critical areas requiring management attention.



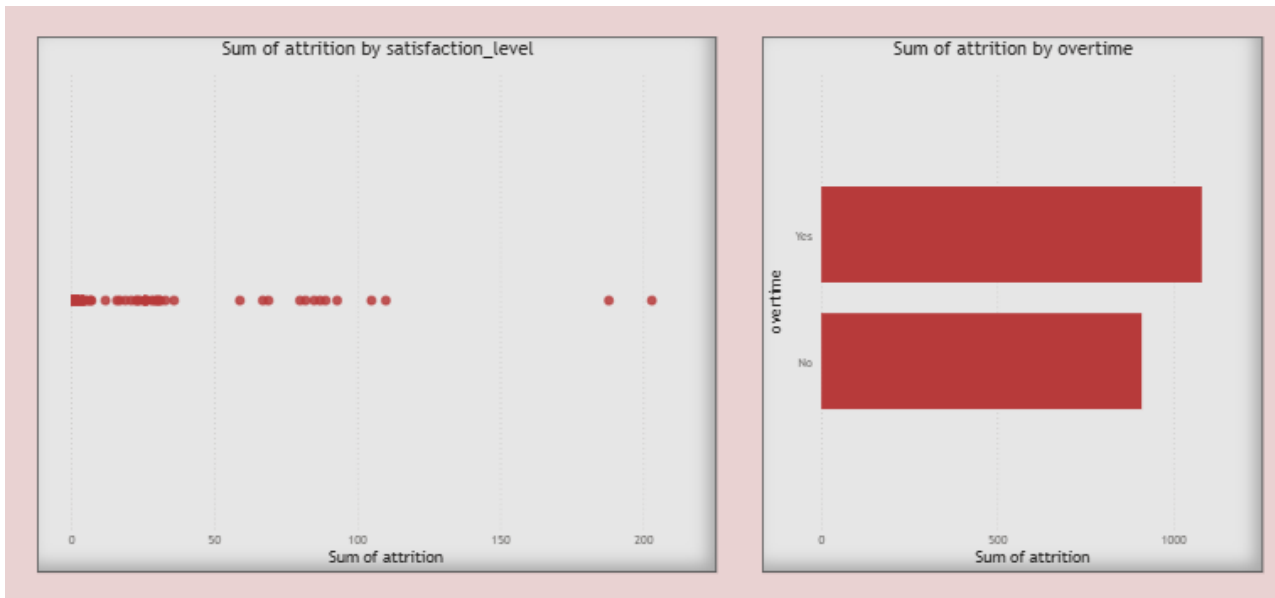
Attrition Analysis Dasshboard:

The Attrition Analysis Dashboard focuses on understanding employee turnover across departments, salary levels, promotion status, and work accident history. By identifying patterns associated with employee exits, the dashboard enables HR professionals to determine which employee groups are more likely to leave the organization and develop targeted retention strategies.



Employee Health Dashboard:

The Employee Health Dashboard evaluates factors related to employee well-being and engagement, including job satisfaction, overtime workload, average monthly working hours, and relationships among workforce variables. This analysis helps assess how workplace conditions influence employee retention and organizational performance.



Employee Segmentation Dashboard:

The Employee Segmentation Dashboard categorizes employees into distinct groups such as High-Risk Employees, Overtime Employees, Long-Term Employees, and Dissatisfied Employees. This segmentation allows HR teams to proactively monitor vulnerable employee groups and implement appropriate interventions before attrition occurs.



BUSINESS QUESTIONS:

1. Which factors contribute most to attrition?

- Low salary
- Low employee satisfaction
- Lack of promotion opportunities
- Overtime workload
- High-risk employee behavior patterns

These factors show the strongest relationship with employee exits.

2. Which departments have the highest employee turnover?

Highest turnover departments:

1. HR (18.80%)
2. Accounting (17.55%)
3. Technical (17.38%)
4. Support (17.13%)

These departments require focused retention initiatives.

3. How does work-life balance affect retention?

Employees working overtime have a higher attrition rate (18.18%) compared to non-overtime employees (15.05%).

This suggests that excessive workload and poor work-life balance contribute to employee exits.

4. Which employees are at highest risk?

High-risk employees show an attrition rate of **50.62%**, much higher than medium-risk (23.66%) and low-risk employees (7.77%).

Typical high-risk employees have:

- Low satisfaction levels
- High working hours
- Overtime workload
- Lower engagement

5. What actions should HR leadership prioritize?

1. Improve employee satisfaction programs.
2. Review compensation for low-salary employees.
3. Reduce excessive overtime.
4. Increase promotion and career growth opportunities.
5. Monitor high-risk employees through workforce analytics dashboards.

6. Focus retention efforts in HR, Accounting, Technical, and Support departments.

KEY FINDINGS:

- The overall attrition analysis revealed that employee satisfaction is one of the most important factors influencing employee retention.
- Employees in the low salary category exhibited higher attrition rates compared to those in medium and high salary groups.
- Overtime employees showed a greater tendency to leave the organization, highlighting the impact of workload and work-life balance on retention.
- Employees who received promotions were less likely to leave, indicating the importance of career growth opportunities.
- The risk segmentation analysis successfully identified high-risk employees who are more likely to leave the organization.

RECOMMENDATIONS:

- Enhance employee satisfaction through engagement programs, recognition initiatives, and regular feedback mechanisms.
- Review compensation policies, particularly for employees in lower salary categories, to improve retention.
- Reduce excessive overtime and encourage a healthy work-life balance to minimize employee burnout.
- Provide clear career development and promotion opportunities to increase employee commitment and motivation.
- Continuously monitor high-risk employees using HR analytics dashboards and implement proactive retention strategies.

INNOVATION:

A rule-based **Employee Attrition Risk Prediction Framework** was developed by introducing a custom **Risk Category** feature. Employees were classified into risk groups based on factors such as job satisfaction and monthly working hours. This approach enables HR teams to proactively identify employees who are at a higher risk of leaving the organization and take preventive actions before attrition occurs.

Benefits:

- Early identification of potential employee exits.
- Supports proactive retention strategies.
- Reduces recruitment and training costs.
- Enhances workforce planning and decision-making.

CONCLUSION:

This project successfully analyzed employee attrition using Python and Power BI to identify the key factors affecting workforce retention. Through data cleaning, exploratory analysis, employee segmentation, and interactive dashboard development, valuable insights were generated regarding employee satisfaction, salary levels, promotions, and overtime. The findings highlight the importance of employee engagement, fair compensation, career growth, and workload management in reducing attrition. The developed dashboard serves as an effective decision-support tool for HR professionals, enabling data-driven strategies to improve employee retention and organizational performance.